

Beta regression – three views

1. Index form

The per-observation reading: what happens for each row i of your data.

$$\begin{aligned}y_i \mid \mu_i, \sigma_i &\sim \text{Beta}(\mu_i \sigma_i, (1 - \mu_i) \sigma_i) \\ \text{logit}(\mu_i) &= \beta_0 + \beta_1 x_i \\ \log(\sigma_i) &= \gamma_0\end{aligned}$$

Coefficient reading. On the response scale, $\exp(\hat{\beta})$ is the odds ratio for the mean proportion: a one-unit increase multiplies the odds $\mu/(1 - \mu)$ by $\exp(\hat{\beta})$ (logit link).

2. Matrix form

The same model in matrix notation. The structural contract every textbook past chapter 4 switches to.

$$\begin{aligned}\mathbf{y} \mid \boldsymbol{\mu}, \boldsymbol{\sigma} &\sim \text{Beta}(\boldsymbol{\mu} \odot \boldsymbol{\sigma}, (1 - \boldsymbol{\mu}) \odot \boldsymbol{\sigma}) \\ \text{logit}(\boldsymbol{\mu}) &= \mathbf{X}\boldsymbol{\beta} \\ \log(\boldsymbol{\sigma}) &= \mathbf{X}_\sigma \boldsymbol{\gamma}\end{aligned}$$

3. Worked observation ($i = 1$)

The index-form equations evaluated at the first observation of your data, with coefficient estimates plugged in.

$$\begin{aligned}\hat{\eta}_1 &= -0.824 - 0.0874 \cdot 1.43 \approx -0.95 \\ \hat{\mu}_1 &= \text{logistic}(\hat{\eta}_1) = \text{logistic}(-0.95) \approx 0.279 \\ y_1 &\sim \text{Beta}(\hat{\mu}_1 \hat{\sigma}_1, (1 - \hat{\mu}_1) \hat{\sigma}_1)\end{aligned}$$

Linear predictor $\hat{\eta}_1 = -0.95$ on the link scale; predicted mean $\hat{\mu}_1 = \text{logistic}(\hat{\eta}_1) \approx 0.279$ on the response scale. No additive error term enters the linear predictor – the spread is set by the likelihood above.

And the σ submodel at $i = 1$:

$$\log \hat{\sigma}_1 = -1.04 \Rightarrow \hat{\sigma}_1 = \exp(-1.04) \approx 0.353$$

Stacking the same response equation for all $n = 100$ observations:

$$\begin{array}{ccc}
\begin{array}{c} \left[\begin{array}{c} -0.95 \\ -0.768 \\ -0.806 \\ -0.79 \\ -0.797 \\ \vdots \\ -0.81 \\ -0.861 \end{array} \right] \\ \underbrace{\hspace{1.5cm}} \end{array} & = & \begin{array}{c} \left[\begin{array}{cc} 1 & 1.43 \\ 1 & -0.651 \\ 1 & -0.207 \\ 1 & -0.393 \\ 1 & -0.32 \\ \vdots & \vdots \\ 1 & -0.164 \\ 1 & 0.421 \end{array} \right] \\ \underbrace{\hspace{1.5cm}} \end{array} \underbrace{\left[\begin{array}{c} -0.824 \\ -0.0874 \end{array} \right]}_{\hat{\beta}_{2 \times 1} \text{ (estimated)}}
\end{array}$$

$\hat{\boldsymbol{\eta}}_{100 \times 1}$ (linear predictor, link scale) $\mathbf{X}_{100 \times 2}$

And the σ submodel (no observed counterpart – σ here is the family's spread parameter and is not a residual SD; see Tab 1 for its specific role in this distribution):

$$\begin{array}{ccc}
\log \begin{array}{c} \left[\begin{array}{c} 0.353 \\ 0.353 \\ 0.353 \\ 0.353 \\ 0.353 \\ \vdots \\ 0.353 \\ 0.353 \end{array} \right] \\ \underbrace{\hspace{1.5cm}} \end{array} & = & \begin{array}{c} \left[\begin{array}{c} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ \vdots \\ 1 \\ 1 \end{array} \right] \\ \underbrace{\hspace{1.5cm}} \end{array} \underbrace{\left[-1.04 \right]}_{\gamma_{1 \times 1}}
\end{array}$$

$\boldsymbol{\sigma}_{100 \times 1}$ $\mathbf{X}_{\sigma, 100 \times 1}$